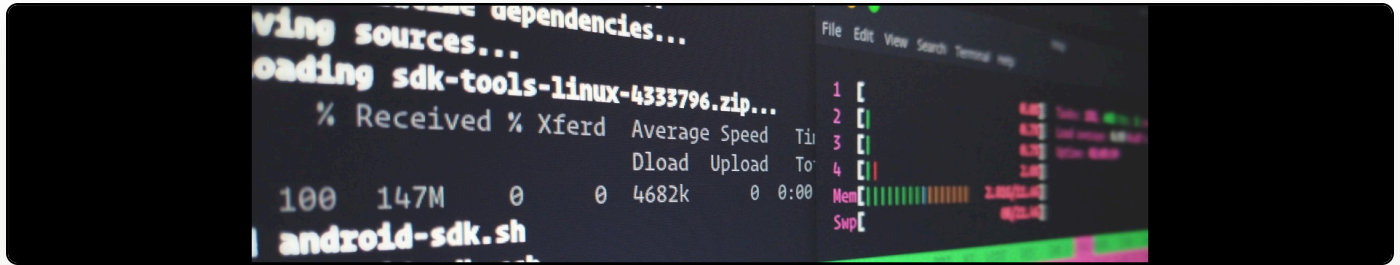


Connecting to AWS OpenSearch Serverless using Python

Cite as: Martin Paul Eve, "Connecting to AWS OpenSearch Serverless using Python", *eve.gd*:
Martin Paul Eve, October 20, 2023, <https://doi.org/10.59348/ay3y8-kk905>.



I recently wanted to use Elasticsearch (or OpenSearch as Amazon terms it from the fork) in an AWS environment, using Python. When I tried to connect I got a very painful 403 Forbidden error. Specifically: `opensearchpy.exceptions.AuthorizationException: AuthorizationException(403, 'Forbidden')`.

The answer I needed was eventually here: <https://opensearch.org/docs/latest/clients/python-low-level/>. The key point is that, when using serverless, the service is not “es” but “aoss” (Amazon OpenSearch Serverless).

My ingest code (that now works) looks like this:

```

def push_to_aws():
    host = f"YOUR_DOMAIN.us-east-1.aoss.amazonaws.com"
    credentials = boto3.Session().get_credentials()
    region = "us-east-1"

    # Sign the credentials
    auth = AWSV4SignerAuth(credentials, region, service="aoss")

    # The OpenSearch client
    client = OpenSearch(
        hosts=[{"host": host, "port": 443}],
        http_auth=auth,
        use_ssl=True,
        verify_certs=True,
        connection_class=RequestsHttpConnection,
    )

    my_index = "gen"

    try:
        response = client.indices.create(my_index)
        print("\nCreating index:")
        print(response)
    except Exception as e:
        print(e)

    files = glob.glob("/home/martin/json_directory/*.json")

    for file in files:
        with open(file, "r") as f:
            print(f"Processing {file}")
            data = payload_constructor(data=json.load(f), action={"index": {}})

            response = client.bulk(body=data, index=my_index)

```